

APPLIED MACHINE LEARNING

- Python
- XGBoost
- Autoencoders
- Anomaly Detection
- Classification
- Time Series
- Forecasting
- Computer Vision

LLM SYSTEMS

- Transformers
- LoRA / QLoRA
- PEFT
- Unsloth
- vLLM
- LLM Evaluation
- Information Extraction
- OpenAI-Compatible APIs

PRODUCTION ML

- GCP
- GKE
- Vertex AI
- Kubernetes
- Seldon Core
- MLflow
- Dataflow / Beam

EVALUATION & RELIABILITY

- Model Monitoring
- Prometheus
- Grafana
- CI/CD
- GitOps
- ArgoCD
- Istio
- Blue-Green Delivery

CERTIFICATIONS

- Oracle Cloud Infrastructure 2024
Generative AI Certified Professional
- Microsoft Certified
Azure AI Engineer Associate
- Microsoft Certified
Azure Data Scientist Associate
- Google Cloud
Certified Generative AI Leader

EDUCATION

- B.E. Electrical & Electronics
Andhra University | 2012 - 2016

ML METHODS

- Scikit-learn
- PyTorch
- TensorFlow
- Keras
- DenseNet
- Predictive Modelling
- Unsupervised Learning
- Feature Engineering
- Transfer Learning
- NLP
- Time-Series Forecasting

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization.

EXPERIENCE

Lead Artificial Intelligence Engineer

Best Buy India
Jul 2025 - Present | Bengaluru

- > Built a config-driven framework for LoRA/QLoRA fine-tuning across classification, information extraction, and generation tasks using Unsloth, PEFT, Transformers, and Accelerate.
- > Automated post-training evaluation with accuracy, confusion-matrix, and MSE outputs; standardized artifacts for hot-loading into a shared vLLM OpenAI-compatible service, with estimated annual API-cost savings of **~\$300K**. Kept model inputs company-managed and reduced external API exposure during peak demand.
- > Built batch, streaming and real-time fraud systems: Beam/Dataflow feature pipelines at **40,000 events/sec at peak**, secure, monitored GKE/Seldon Core inference and load testing. ML feeds a rules-based engine; combined fraud prevention saves **US\$1.5M/year**.
- > Owned model rollout, HPA scaling, Istio routing, blue-green delivery, Cloud Build, ArgoCD, Prometheus, Grafana, and production load testing.

Lead ML Engineer — Machine Learning & MLOps

Wipro
Aug 2023 - Jul 2025 | Hyderabad

- > Built continuously retrained behavioral autoencoders on NVIDIA DGX (8xA100), processing **billion-row** AWS CloudTrail datasets with Morpheus, RAPIDS, MLflow, and Grafana monitoring.
- > Delivered phishing classification at **2,400 emails/sec** - 2x prior throughput - with feedback-aware training and inference pipelines integrated with Microsoft Sentinel.
- > Engineered multifomat abuse-inbox classification and information extraction across text, HTML, images, PDF, PNG, and JPEG using Python, OCR, and GPU-accelerated processing.
- > Led ML/MLOps delivery and mentoring at Wipro; completed **300+ résumé screens** and **50+ technical interviews**.

Senior Data Scientist — Data Science / Azure ML

Accenture
May 2021 - Sep 2023 | India

- > Led a production quality-prediction system with automated retraining, monitoring, and stakeholder delivery; increased inspections from **130 to 160 trucks/day**, improved low-quality detection 17%, and generated **\$2.5M annual savings**.
- > Built applied ML and optimization systems using Python, Azure ML, Azure Functions, Logic Apps, CI/CD, and heterogeneous operational data.

PROJECTS

LLM Training, Evaluation & Serving
Best Buy | LoRA | vLLM | GKE

A shared path from task configuration and fine-tuning through evaluation, artifact handling, autoscaled serving, controlled rollout, and production observability.

Behavioral Anomaly Detection
Wipro | DGX | 1B+ rows

Per-entity autoencoder models, continuously retrained on high-cardinality CloudTrail data and monitored through operational security dashboards.

GPU STACK

CUDA GPU Libraries

RAPIDS cuDF / CuPy

Multi-GPU Training

OPTIMISATION & SIMULATION

Google OR-Tools

Vehicle Routing

Mixed-Integer Programming

Constraint Modelling

Operations Research

Assignment Models

Supply-Chain Optimization

Scenario Analysis

CLOUD & MANAGED ML

GCS

Artifact Registry

Azure ML

Azure DevOps

Function Apps

Logic Apps

Cloud Build

Google KCC

DATA & ML LIFECYCLE

Apache Beam

Dataflow

Pub/Sub

Vertex AI Feature Store

Streaming Pipelines

Training Pipelines

Automated Evaluation

Experiment Tracking

Model Registry

Model Versioning

Continuous Training

LANGUAGES & DATA

SQL

Pandas

NumPy

MS SQL Server

IBM DB2

Power BI

ENGINEERING LEADERSHIP

Architecture Reviews

Technical Mentoring

Technical Hiring

Team Building

Stakeholder Alignment

Cross-Functional Delivery

ACHIEVEMENTS

15 honors and awards across Best Buy, Wipro, Accenture, and Infosys.
Best Buy Talent Quotient (2026): cybersecurity and fraud delivery.
Best Buy client AML Hall of Fame (2024): DGX delivery during the Wipro engagement.

ADDITIONAL PROJECTS & EXPERIENCE

Document Classification & Extraction
Wipro | OCR | Python

Classification, OCR, normalization, and identifier extraction across noisy email bodies and attachments, routed into real security-response workflows.

Forecasting & Quality ML
GlobalData / Accenture | Production ML

Domain-aware forecasting across 15+ verticals and monitored quality prediction with automated retraining and measurable operational outcomes.

EARLIER EXPERIENCE & ACHIEVEMENTS

Data Scientist
GD Research Centre | May 2020 - May 2021

Built domain-aware time-series forecasting and scenario analysis across energy, manufacturing, healthcare, finance, retail, telecommunications, automotive, and other industry intelligence centers.

Deep Learning Engineer
Tata Consultancy Services | Jul 2019 - Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on **~1 million images in ~3 hours**, with **90% classification accuracy**. Deployed at Amkor, South Korea, and trained the client team.

Senior Software Engineer - ML
Infosys | May 2016 - Jul 2019

Built an ML-powered lead-conversion prediction system and business-intelligence applications that moved sales prioritisation from intuition to data.

Achievements
15 honors and awards

Best Buy Talent Quotient (2026); Best Buy client AML Hall of Fame (2024), during the Wipro engagement.

Vehicle Routing & Transport Planning
Accenture | Google OR-Tools | Operations research

Designed vehicle-routing and mixed-integer models covering capacity, time windows, waiting time, demand, and departure constraints, then automated daily operational planning.

Industry Forecasting Platform
GD Research Centre | Time series | Scenario analysis

Designed and deployed domain-aware forecasting across 15+ industries, combining feature engineering, time-series modelling, and scenario analysis for enterprise decisions.

System Ownership
Design | Implementation | Production operations | Maintenance

Owned AI/ML systems from data and training pipelines through evaluation, deployment, monitoring, maintenance, incident support, and continuous improvement.

Engineering Leadership & Team Building
Architecture | Mentoring | Hiring | Cross-functional delivery

Stayed hands-on while leading ML/MLOps delivery, mentoring engineers, coordinating client and platform stakeholders, and helping scale the India AI engineering hub during the Wipro engagement through **300+ résumé screens** and **50+ technical interviews**.